
Beginner Level Terraform Assessments

□ 1. Launch an EC2 Instance (Free Tier)

□ Goal:

Use Terraform to launch a basic virtual server (EC2) on AWS using the free tier.

□ What you'll do:

- Set up AWS as the cloud provider in Terraform
- Use the `aws_instance` resource to create an EC2 instance

- Use your own .pem key manually (to SSH into the instance)

□ Expected Result:

- The EC2 instance is visible in the AWS Console
- You can connect to it using SSH
- Terraform should also show its public IP as output

□ 2. Create an S3 Bucket

□ Goal:

Use Terraform to create a cloud storage bucket (S3).

□ What you'll do:

- Use `aws_s3_bucket` resource
- Make sure the bucket name is globally unique
- Enable versioning so older versions of files are saved

□ Expected Result:

- S3 bucket is created and visible in AWS
- Versioning is enabled in settings

□ 3. Create a Security Group

□ Goal:

Create a firewall (called a security group in AWS) that allows incoming traffic for:

- SSH (port 22)
- HTTP (port 80)

□ What you'll do:

- Use `aws_security_group` resource
- Add rules to allow SSH and HTTP access from your IP or anywhere

□ Expected Result:

- When applied to EC2, you can access it via SSH and from a browser
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□ 4. Use Variables and Outputs

□ Goal:

Make your Terraform code more dynamic and reusable.

□ What you'll do:

- Create variables.tf to define inputs (like instance type, region, AMI ID)
- Use terraform.tfvars to store actual values

- Create outputs.tf to show values like EC2's public IP after deployment

□ Expected Result:

- Your code becomes cleaner and easier to update
- Terraform shows the EC2 IP or name after deployment

□ Sample Folder Structure for Beginner Projects

terraform-ec2-setup/

- main.tf □ Main Terraform code
- variables.tf □ All input variables
- outputs.tf □ Show values like public IP
- terraform.tfvars □ Actual values for variables

provider.tf AWS credentials and region

Tools You Need (Before You Start):

Tool	Purpose
Terraform	To write and run infrastructure as code
AWS CLI	To configure your AWS access
